

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 1

Analytical Problem Solving

COURSE NAME: Deep Learning

COURSE CODE: MLA0403

FACULTY : Dr.Arul Raj A.M

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I Aryan.KS / 192325051 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name of the student: Aryan.KS

A handwritten signature in black ink on a light gray background. The signature is cursive and appears to read 'Aryan'.

QUESTIONS:05

Total Marks:25

Analytical Problem Solving

1. Learning Algorithms (Optimization Behavior)

A machine learning model is trained using gradient descent on a convex loss function. Initially, the learning rate is set very high, and later it is gradually decreased.

Question:

Analyze how the choice of a high initial learning rate followed by decay affects convergence. Under what conditions can this strategy outperform a constant learning rate? Discuss possible risks and justify with reasoning.

ANS:

A high initial learning rate helps the model converge faster by making large updates. Gradually decreasing the learning rate allows fine-tuning near the optimum. This strategy outperforms a constant learning rate when the loss function is smooth and convex. Risks include overshooting the minimum, unstable training, or divergence if the initial learning rate is too high.

2. Maximum Likelihood Estimation (MLE)

Suppose you are given a dataset generated from a Gaussian distribution with unknown mean μ and known variance σ^2 .

Question:

Derive the Maximum Likelihood Estimator for μ , and analytically explain how the estimate changes when:

- The dataset contains outliers
 - The sample size increases
- What does this imply about the robustness of MLE?

ANS:

Maximum Likelihood Estimation (MLE)

The likelihood is:

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

The MLE of the mean is:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$

- **With outliers:** The estimate shifts toward the outliers because the mean is sensitive to extreme values.
- **As sample size increases:** The estimate becomes more accurate and converges to the true mean.

3. Building a Machine Learning Algorithm (Model Selection)

You are designing a classification algorithm for a dataset with 10,000 features but only 500 training samples.

Question:

Analyze the challenges involved in training such a model. Propose a strategy (algorithm pipeline) to handle this situation and justify each step (e.g., feature selection, regularization, dimensionality reduction). Explain how your approach mitigates overfitting.

ANS:

Challenges: High-dimensional data with few samples causes overfitting and high computation.

Pipeline:

1. Feature selection to remove irrelevant features.
2. Dimensionality reduction (PCA) to reduce feature space.
3. Regularization (L1/L2) to control model complexity.
4. Cross-validation for model selection.

4. Neural Networks (Multilayer Perceptron - MLP)

Consider a Multilayer Perceptron with one hidden layer using sigmoid activation functions.

Question:

Explain analytically why adding more hidden neurons increases the representational power of the network. Then, reason why this does not always lead to better performance. Support your answer using concepts like overfitting and generalization.

ANS:

Adding more hidden neurons increases the network's ability to learn complex patterns and nonlinear relationships. However, too many neurons increase model complexity, leading to overfitting and poor generalization on unseen data. An optimal number of neurons balances learning capacity and generalization.

5. Backpropagation, SGD & Curse of Dimensionality

A deep neural network is trained using Stochastic Gradient Descent (SGD) on a very high-dimensional dataset.

Question:

Analyze how the **curse of dimensionality** impacts:

- Gradient updates in backpropagation
- Convergence speed of SGD
- Model generalization

Propose at least two techniques to overcome these issues and justify them analytically.

ANS:

Backpropagation, SGD & Curse of Dimensionality

- **Gradient updates:** Become noisy and less informative in high dimensions.
- **SGD convergence:** Slower due to sparse data and complex optimization.
- **Generalization:** Higher risk of overfitting because of many features.

Techniques:

1. **Dimensionality reduction (PCA):** Removes redundant features and speeds up training.
2. **Regularization (L2/Dropout):** Reduces overfitting and improves generalization